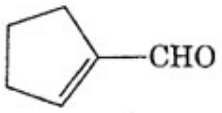
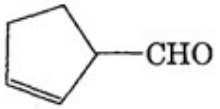
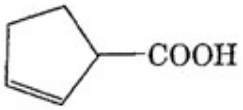
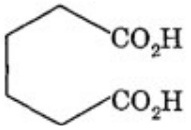
26. Cyclohexene on ozonolysis followed by reaction with zinc dust and water gives compound **E**. Compound **E** on further treatment with aqueous KOH yields compound **F**. Compound **F** is

(A) 

(B) 

(C) 

(D) 

Answer ● ○ ○ ○

(A) (B) (C) (D)

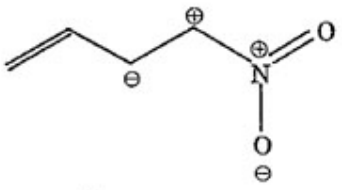
27. The number of stereoisomers obtained by bromination of *trans*-2-butene is

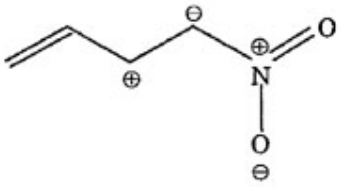
(A) 1 (B) 2 (C) 3 (D) 4

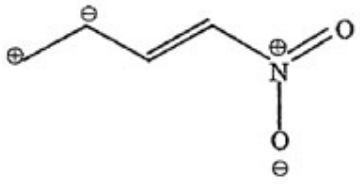
Answer ● ○ ○ ○

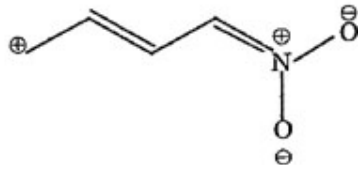
(A) (B) (C) (D)

28. Among the following, the least stable resonance structure is

(A) 

(B) 

(C) 

(D) 

Answer ● ○ ○ ○

(A) (B) (C) (D)

29. A positron is emitted from $^{23}_{11}Na$. The ratio of the atomic mass and atomic number of the resulting nuclide is

(A) 22/10 (B) 22/11

(C) 23/10 (D) 23/12

Answer ○ ○ ● ○

(A) (B) (C) (D)

30. For the process $H_2O(l)$ (1 bar, 373 K) → $H_2O(g)$ (1 bar, 373 K), the correct set of thermodynamic parameters is

(A) $\Delta G = 0$, $\Delta S = +ve$ (B) $\Delta G = 0$, $\Delta S = -ve$

(C) $\Delta G = +ve$, $\Delta S = 0$ (D) $\Delta G = -ve$, $\Delta S = +ve$

Answer ● ○ ○ ○

(A) (B) (C) (D)

31. Consider a reaction aG + bH → Products. When concentration of both the reactants G and H is doubled, the rate increases by eight times. However, when concentration of G is doubled keeping the concentration of H fixed, the rate is doubled. The overall order of the reaction is

(A) 0 (B) 1 (C) 2 (D) 3

Answer ○ ○ ○ ●

(A) (B) (C) (D)

32. STATEMENT-1 : Alkali metals dissolve in liquid ammonia to give blue solutions.

because

STATEMENT-2 : Alkali metals in liquid ammonia give solvated species of the type $[M(NH_3)_n]^+$ (M = alkali metals).

(A) Statement-1 is True, Statement-2 is True; Statement-2 **is** a correct explanation for Statement-1

(B) Statement-1 is True; Statement-2 is True; Statement-2 **is NOT** a correct explanation for Statement-1

(C) Statement-1 is True, Statement-2 is False

(D) Statement-1 is False, Statement-2 is True

Answer ○ ● ○ ○

(A) (B) (C) (D)

33. STATEMENT-1 : Glucose gives a reddish-brown precipitate with Fehling's solution.

because

STATEMENT-2 : Reaction of glucose with Fehling's solution gives CuO and gluconic acid.

(A) Statement-1 is True, Statement-2 is True; Statement-2 **is** a correct explanation for Statement-1

(B) Statement-1 is True, Statement-2 is True; Statement-2 **is NOT** a correct explanation for Statement-1

(C) Statement-1 is True, Statement-2 is False

(D) Statement-1 is False, Statement-2 is True

Answer ○ ○ ● ○

(A) (B) (C) (D)

34. STATEMENT-1 : Molecules that are not superimposable on their mirror images are chiral.

because

STATEMENT-2 : All chiral molecules have chiral centres.

(A) Statement-1 is True, Statement-2 is True; Statement-2 **is** a correct explanation for Statement-1

(B) Statement-1 is True, Statement-2 is True; Statement-2 **is NOT** a correct explanation for Statement-1

(C) Statement-1 is True, Statement-2 is False

(D) Statement-1 is False, Statement-2 is True

Answer ○ ○ ● ○

(A) (B) (C) (D)

35. STATEMENT-1 : Band gap in germanium is small.

because

STATEMENT-2 : The energy spread of each germanium atomic energy level is infinitesimally small.

(A) Statement-1 is True, Statement-2 is True; Statement-2 **is** a correct explanation for Statement-1

(B) Statement-1 is True, Statement-2 is True; Statement-2 **is NOT** a correct explanation for Statement-1

(C) Statement-1 is True, Statement-2 is False

(D) Statement-1 is False, Statement-2 is True

Answer ○ ● ○ ○

(A) (B) (C) (D)

36. Among the following, identify the correct statement.

(A) Chloride ion is oxidised by O_2 (B) Fe^{2+} is oxidised by iodine

(C) Iodide ion is oxidised by chlorine (D) Mn^{2+} is oxidised by chlorine

Answer ○ ○ ● ○

(A) (B) (C) (D)

37. While Fe^{3+} is stable, Mn^{3+} is not stable in acid solution because

(A) O_2 oxidises Mn^{2+} to Mn^{3+}

(B) O_2 oxidises both Mn^{2+} to Mn^{3+} and Fe^{2+} to Fe^{3+}

(C) Fe^{3+} oxidises H_2O to O_2

(D) Mn^{3+} oxidises H_2O to O_2

Answer ○ ○ ○ ●

(A) (B) (C) (D)

38. Sodium fusion extract, obtained from aniline, on treatment with iron(II) sulphate and H_2SO_4 in presence of air gives a Prussian blue precipitate. The blue colour is due to the formation of

(A) $Fe_4[Fe(CN)_6]_3$ (B) $Fe_3[Fe(CN)_6]_2$

(C) $Fe_4[Fe(CN)_6]_2$ (D) $Fe_3[Fe(CN)_6]_3$

Answer ● ○ ○ ○

(A) (B) (C) (D)

39. Which one of the following reagents is used in the above reaction?

(A) aq. NaOH + CH_3Cl (B) aq. NaOH + CH_2Cl_2

(C) aq. NaOH + $CHCl_3$ (D) aq. NaOH + CCl_4

Answer ○ ○ ● ○

(A) (B) (C) (D)

10. The electrophile in this reaction is

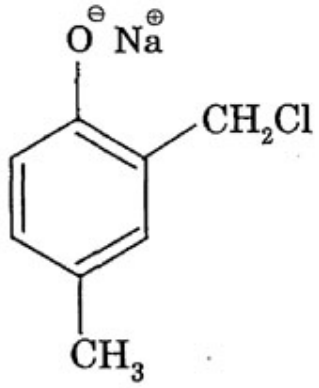
(A) :CHCl (B) $^{+}CHCl_2$ (C) $:CCl_2$ (D) $\cdot CCl_3$

Answer ○ ○ ● ○

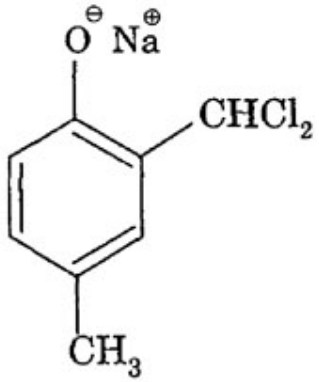
(A) (B) (C) (D)

41. The structure of the intermediate **I** is

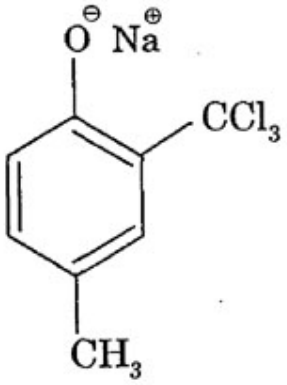
(A)



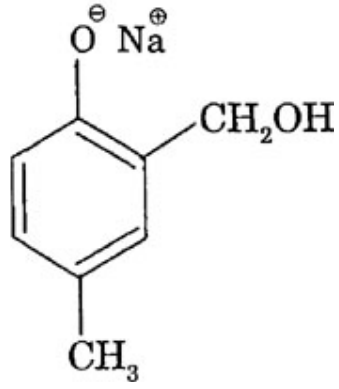
(B)



(C)



(D)



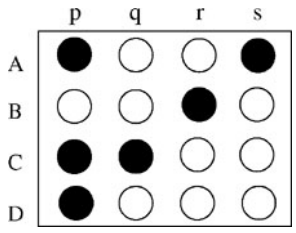
Answer ○ ● ○ ○

(A) (B) (C) (D)

42. Match the reactions in **Column I** with nature of the reactions/type of the products in **Column II**. Indicate your answer by darkening the appropriate bubbles of the 4×4 matrix given in the ORS.

Column I	Column II
(A) $O_2^- \rightarrow O_2 + O_2^{2-}$	(p) redox reaction
(B) $CrO_4^{2-} + H^+ \rightarrow$	(q) one of the products has trigonal planar structure
(C) $MnO_4^- + NO_2^- + H^+ \rightarrow$	(r) dimeric bridged tetrahedral metal ion
(D) $NO_3^- + H_2SO_4 + Fe^{2+} \rightarrow$	(s) disproportionation

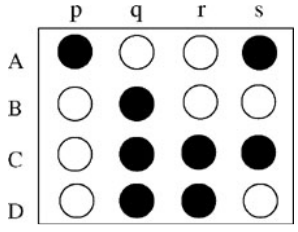
Answer



43. Match the compounds/ions in **Column I** with their properties/reactions in **Column II**. Indicate your answer by darkening the appropriate bubbles of the 4×4 matrix given in the ORS.

Column I	Column II
(A) C_6H_5CHO	(p) gives precipitate with 2,4-dinitrophenylhydrazine
(B) $CH_3C \equiv CH$	(q) gives precipitate with $AgNO_3$
(C) CN^-	(r) is a nucleophile
(D) I^-	(s) is involved in cyanohydrin formation

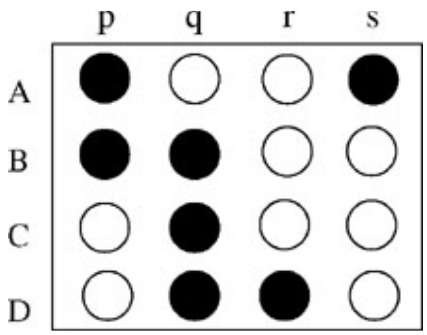
Answer



44. Match the crystal system/unit cells mentioned in **Column I** with their characteristic features mentioned in **Column II**. Indicate your answer by darkening the appropriate bubbles of the 4×4 matrix given in the ORS.

Column I	**Column II**
(A) simple cubic and face-centred cubic	(p) have these cell parameters $a=b=c$ and $\alpha=\beta=\gamma$
(B) cubic and rhombohedral	(q) are two crystal systems
(C) cubic and tetragonal	(r) have only two crystallographic angles of 90°
(D) hexagonal and monoclinic	(s) belong to same crystal system

Answer



45. Let $O(0, 0)$, $P(3, 4)$, $Q(6, 0)$ be the vertices of the triangle OPQ. The point R inside the triangle OPQ is such that the triangles OPR, PQR, OQR are of equal area. The coordinates of R are

(A) $\left(\frac{4}{3}, 3\right)$ (B) $\left(3, \frac{2}{3}\right)$

(C) $\left(3, \frac{4}{3}\right)$ (D) $\left(\frac{4}{3}, \frac{2}{3}\right)$

Answer



(A) (B) (C) (D)